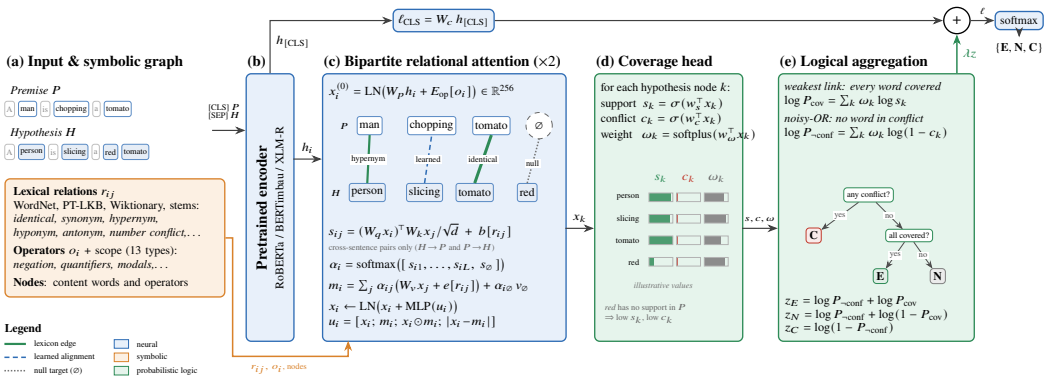
Experimentos em Natural Language Inference (NLI)

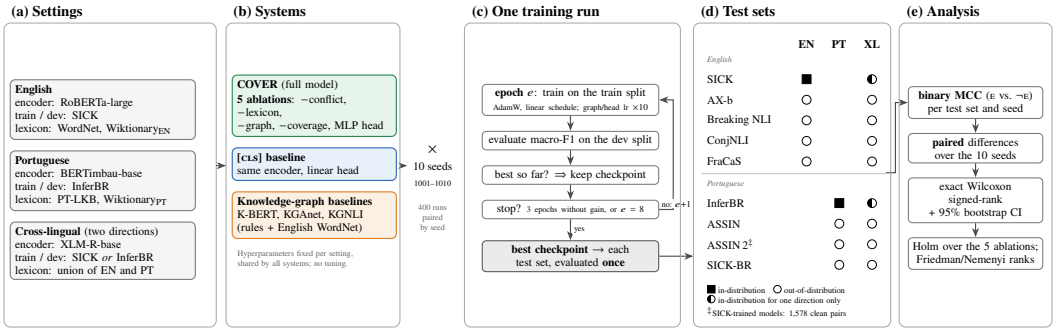
17 de novembro de 2026

COVER: a ideia



Para cada palavra da hipótese: **suporte** (a premissa cobre?) e **conflito** (a premissa contradiz?) ⇒ regra lógica: E = tudo coberto e sem conflito; N = algo não coberto; C = algum conflito.

Desenho das experimentações



Hiperparâmetros fixos por configuração, sem ajuste por variante; tudo pareado por seed.

Conjuntos de teste

Conjunto	Língua	Pares	Testa
SICK	EN	4.906	in-distribution
AX-b	EN	1.104	fenômenos linguísticos
Breaking NLI	EN	8.193	troca de 1 palavra
ConjNLI	EN	623	conjunções
FraCaS	EN	335	semântica formal
InferBR	PT	1.705	in-distribution
ASSIN	PT	4.000	OOD
ASSIN 2	PT	2.448	OOD
SICK-BR	PT	4.906	tradução do SICK

- Em negrito: conjuntos de desafio adicionados.
- HANS e MED foram testados e retirados (tempo de avaliação).
- SICK-BR é alinhado par a par com o SICK.
- **Métricas**: acurácia e MCC da decisão binária E vs. $\neg\mathrm{E}$, as únicas definidas em todos os conjuntos.

Resultados principais — inglês

(a) English: RoBERTa-large trained on SICK

† $p < .05$ contra [CLS]

	Accuracy					MCC				
System	SICK	AX-b	Break.	Conj.	FraCaS	SICK	AX-b	Break.	Conj.	FraCaS
[CLS]	94.3 $\pm$0.4	69.1 $\pm$2.5	**95.2** $\pm$1.1	63.0 $\pm$6.2	59.6 $\pm$3.3	86.0 $\pm$0.9	38.2 $\pm$3.7	80.3 $\pm$3.0	34.2 $\pm$4.5	19.6 $\pm$3.8
K-BERT	93.3† $\pm$0.4	66.5† $\pm$1.2	95.2 $\pm$0.7	57.6 $\pm$4.6	58.7 $\pm$1.4	83.9† $\pm$0.9	33.4† $\pm$1.4	**80.3** $\pm$2.8	22.6† $\pm$4.2	16.3† $\pm$1.9
KGAnet	94.3 $\pm$0.4	63.0† $\pm$3.1	93.4† $\pm$2.0	53.2† $\pm$5.1	62.8 $\pm$2.2	86.1 $\pm$0.8	28.9† $\pm$4.7	75.1† $\pm$6.0	21.3† $\pm$4.1	20.6 $\pm$3.7
KGNLI	**94.4** $\pm$0.2	61.4† $\pm$2.1	93.0† $\pm$1.5	51.6† $\pm$3.3	63.9† $\pm$1.3	**86.3** $\pm$0.5	26.4† $\pm$3.0	73.8† $\pm$3.9	19.0† $\pm$3.4	21.4 $\pm$3.3
COVER	94.2 $\pm$0.2	**71.6**† $\pm$1.5	95.0 $\pm$1.3	**66.7** $\pm$4.7	**65.7**† $\pm$2.6	85.9 $\pm$0.6	**43.1**† $\pm$2.3	79.6 $\pm$4.2	**40.1**† $\pm$3.9	**28.2**† $\pm$3.3

SICK: **empate**. Fora da distribuição: +4.9 AX-b, +5.9 ConjNLI, +8.6 FraCaS (MCC). Breaking NLI: empate. **Baselines de grafo**: empatam (ou quase) no SICK mas caem fora dele; COVER supera KGAnet, KGNLI e K-BERT em +10 a +21 MCC em AX-b e ConjNLI (10/10 seeds).

Resultados principais — português

(b) Portuguese: BERTimbau-base trained on InferBR

† $p < .05$ contra [CLS]

	Accuracy				MCC			
System	InferBR	ASSIN	ASSIN 2	SICK-BR	InferBR	ASSIN	ASSIN 2	SICK-BR
[CLS]	95.2 ±0.5	82.1 ±2.1	83.9 ±1.6	82.5 ±1.5	89.2 ±1.2	57.2 ±3.4	68.0 ±3.0	58.7 ±3.2
K-BERT								
KGAnet								
KGNLI								
COVER	**95.7** ±0.5	**83.1** ±2.0	**86.3**† ±0.4	**84.6**† ±0.5	**90.4** ±1.0	**60.5**† ±3.7	**72.6**† ±0.9	**64.0**† ±0.8

InferBR: **empate** (+1.2 MCC, n.s.). Fora da distribuição: **+3.3** ASSIN, **+4.6** ASSIN 2, **+5.3** SICK-BR (MCC), com ∼1/3 do desvio entre seeds.

Transferência entre línguas

(c) Cross-lingual EN→PT: XLM-R-base trained on SICK, Portuguese tests

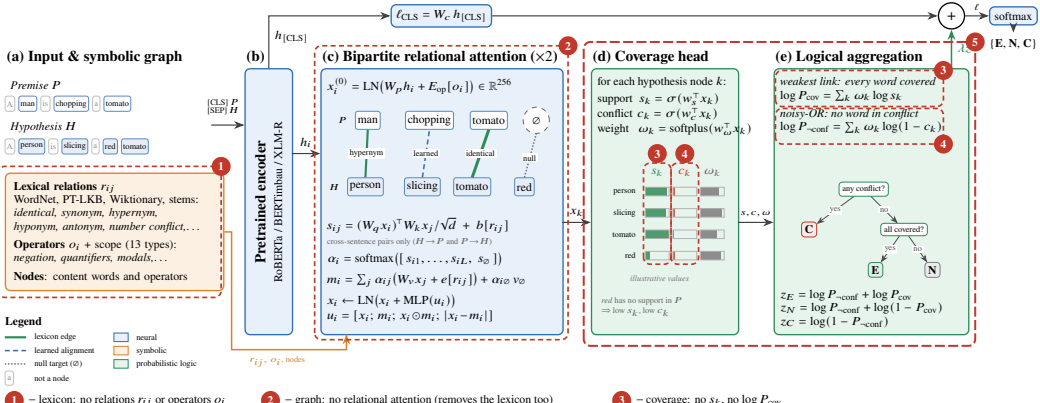
	Accuracy				MCC			
System	InferBR	ASSIN	ASSIN 2	SICK-BR	InferBR	ASSIN	ASSIN 2	SICK-BR
[CLS]	**79.2** ±1.5	77.3 ±6.7	86.3 ±1.7	**86.5** ±0.6	51.8 ±3.5	47.4 ±8.7	70.9 ±3.5	**66.4** ±1.3
K-BERT	65.5† ±4.1	69.1† ±2.9	74.9† ±3.7	74.8† ±4.0	28.1† ±4.7	25.8† ±3.6	48.8† ±6.4	46.1† ±5.7
KGAnet	66.0† ±2.5	**80.1** ±3.1	72.5† ±1.7	73.4† ±0.6	25.6† ±4.3	**48.0** ±5.2	41.9† ±4.2	38.7† ±2.0
KGNLI								
COVER	78.7 ±1.7	77.0 ±5.2	**86.6** ±0.6	86.0 ±0.5	**52.6** ±2.7	46.1 ±5.4	**71.7** ±1.4	65.8 ±1.2

(d) Cross-lingual PT→EN: XLM-R-base trained on InferBR, English tests

	Accuracy					MCC				
System	SICK	AX-b	Break.	Conj.	FraCaS	SICK	AX-b	Break.	Conj.	FraCaS
[CLS]	78.7 ±1.4	60.9 ±0.9	**87.8** ±1.5	69.3 ±2.4	58.2 ±1.3	52.6 ±2.8	18.0 ±1.9	**44.5** ±5.9	28.4 ±6.7	11.3 ±2.8
K-BERT										
KGAnet										
KGNLI										
COVER	**80.9**† ±0.9	**61.3** ±1.5	86.2 ±3.4	**70.3** ±2.0	**60.3**† ±1.7	**57.4**† ±1.3	**19.2** ±3.8	42.6 ±6.9	**34.7**† ±3.5	**13.6** ±3.0

EN→PT: **empate** com o [CLS] nos testes em português (ASSIN 2 só nos pares que não vêm do treino do SICK); K-BERT e KGAnet ficam ∼20–27 MCC abaixo no InferBR e no SICK-BR (KGNLI em execução). PT→EN (10 seeds): COVER ganha no SICK **+4.7** e no ConjNLI **+6.3**; AX-b e FraCaS positivos mas n.s.; Breaking NLI empate. Confirma a direção promissora dos testes preliminares.

Ablação: o que cada variante remove



1 – lexicon: no relations r_{ij} or operators o_i

2 – graph: no relational attention (removes the lexicon too)

3 – coverage: no s_k, no log P_{cov}

4 – conflict: no c_k, no log $P_{\neg\mathrm{conf}}$

5 MLP head: panels (d) and (e) replaced by an MLP over the hypothesis nodes x_k

Ablação (Δ MCC contra o COVER completo)

(a) English: RoBERTa-large trained on SICK

System	SICK	AX-b	Break.	Conj.	FraCaS
COVER (full)	85.9 ±0.6	43.1 ±2.3	79.6 ±4.2	40.1 ±3.9	28.2 ±3.3
− conflict	+0.2	+2.0	+2.8	+0.4	−0.2
− lexicon	−0.4	+0.2	+0.7	−1.6	+0.5
− graph	+0.2	−1.8	+1.1	−4.8	−0.3
− coverage	+0.2	−6.7*	+0.4	−8.5*	−9.6*
MLP head	−0.2	−5.1*	−3.2	−11.8*	−2.0
[CLS]	+0.1	−4.9†	+0.7	−5.9†	−8.6†

(b) Portuguese: BERTimbau-base trained on InferBR

System	InferBR	ASSIN	ASSIN 2	SICK-BR
COVER (full)	90.4 ±1.0	60.5 ±3.7	72.6 ±0.9	64.0 ±0.8
− conflict	−0.1	+2.4	+1.1*	+1.0*
− lexicon	−0.9	+1.5	−1.5*	−1.3*
− graph	−0.6	−0.3	−2.1*	−1.8*
− coverage	−0.6	−11.9*	−4.0*	−4.2*
MLP head	−0.1	−2.4	−1.4	−0.8
[CLS]	−1.2	−3.3†	−4.6†	−5.3†

* **Holm** $p < .05$ **contra o completo**; † $p < .05$.

Ablação na transferência EN→PT (XLM-R treinado em SICK)

Testes em português; Δ MCC contra o COVER completo *Holm $p < .05$; † $p < .05$ contra o completo

System	InferBR	ASSIN	ASSIN 2	SICK-BR
COVER (full)	52.6 ±2.7	46.1 ±5.4	71.7 ±1.4	65.8 ±1.2
− conflict	−2.4	+0.2	−0.5	+0.3
− lexicon	−1.9	+1.5	+1.3	+2.1*
− graph	+0.9	+1.2	+1.0	+2.1*
− coverage	−1.0	−0.9	−1.4	+0.9
MLP head	+2.1	+2.1	+0.6	+0.6
[CLS]	−0.8	+1.3	−0.8	+0.7

Ablação completa. Na transferência para o português **nenhum componente ajuda**: nenhuma variante é significativamente pior que o COVER completo, e tirar o grafo ou o léxico até melhora o SICK-BR (+2.1, Holm). As relações aprendidas em inglês não parecem transferir para as arestas em português. PT→EN em execução.

Ablação: o que aprendemos

Componente	Inglês (RoBERTa-large)	Português (BERTimbau)	EN→PT (XLM-R)
Cobertura	**essencial**: sem ela AX-b −6.7, Conj. −8.5, FraCaS −9.6	**essencial**: ASSIN −11.9, ASSIN 2 −4.0, SICK-BR −4.2	n.s.
Regra lógica	**essencial**: MLP head AX-b −5.1, Conj. −11.8	secundária: MLP head −0.8 a −2.4 (n.s.)	n.s.
Grafo	ajuda no ConjNLI (−4.8)	ajuda fora da distribuição (−1.8 a −2.1)	**atrapalha** (sem ele: SICK-BR +2.1)
Léxico	não faz diferença	ajuda (−1.3 a −1.5)	**atrapalha** (sem ele: SICK-BR +2.1)
Conflito	**dispensável** (sem ele: +2.0 AX-b)	**dispensável** (sem ele: +1.0 a +2.4)	n.s.

O ganho vem da **cobertura**; o conflito não ajuda em nenhuma configuração (nos testes preliminares em português parecia ajudar). Na transferência EN→PT nenhum componente ajuda. Pergunta em aberto: retirar o conflito do modelo principal?

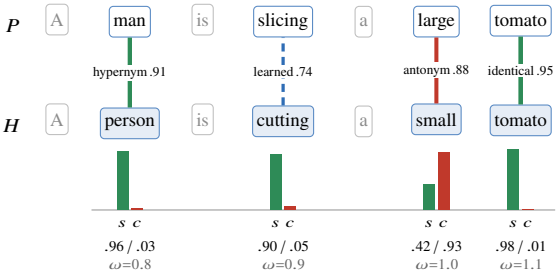
Análise de erros: AX-b por fenômeno (inglês, MCC)

Categoria	n	[CLS]	COVER	Δ
Lexical semantics	368	47.7	50.2	**+2.5**
Predicate–argument	424	45.4	51.7	**+6.3**
Logic	364	19.9	26.2	**+6.3**
Knowledge	284	29.5	31.5	+2.0

Fenômeno	n	Δ
Genitives/Partitives	20	**+33.1**
Nominalization	28	**+28.2**
Morphological negation	26	**+24.3**
Temporal	32	**+24.1**
Double negation	28	**+22.5**
Restrictivity	26	**−15.8**
Universal	18	**−10.1**
Symmetry/Collectivity	28	**−8.7**

Ganho concentrado em estrutura e lógica, não em conhecimento lexical. Fenômenos com $n < 40$ são exploratórios (tabela completa no artigo).

Interpretabilidade



$\log P_{\text{cov}} = \sum_j \omega_j \log s_j = -1.02 \quad (P_{\text{cov}} = .36)$
$\log P_{\neg\text{conf}} = \sum_j \omega_j \log(1 - c_j) = -2.74 \quad (P_{\neg\text{conf}} = .07)$
$p(\text{E}, \text{N}, \text{C}) = \text{softmax}(z) = (.02,\ .04,\ \mathbf{.94}) \Rightarrow$ **contradiction**

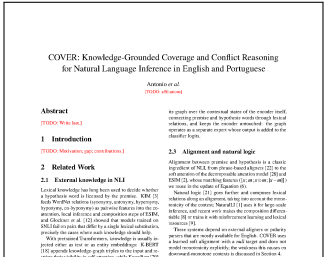
valores ilustrativos

- Cada decisão se lê palavra por palavra: quem cobre, quem contradiz.
- A parte estruturada é quem decide de fato: sozinha, MCC **.84** no SICK contra **.45** da parte [CLS].
- Ela muda 29% das decisões do [CLS]: corrige 25%, estraga 3%.

Conclusões até agora

1. **Padrão robusto:** COVER empata no conjunto de treino e ganha fora dele, em inglês e em português, com três encoders.
2. **O mecanismo é a cobertura**; a regra lógica pesa no inglês; grafo e léxico ajudam no português.
3. **O conflito não ajuda** (diferente dos testes preliminares em português).
4. COVER tem **menor variância entre seeds** (ex.: SICK-BR ± 0.8 vs. ± 3.2).
5. **Cross-lingual:** EN$\rightarrow$PT não transfere; PT$\rightarrow$EN transfere (SICK **+4.7**, ConjNLI **+6.3**).
6. **Baselines de grafo** (KGAnet, KGNLI): empatam no SICK, mas ficam abaixo até do [CLS] fora da distribuição.
7. Correção metodológica: 36% do teste do ASSIN 2 vem do treino do SICK; modelos treinados em SICK são avaliados só no restante.

Início da escrita do artigo



COVER: Knowledge-Grounded Coverage and Conflict Reasoning for Natural Language Inference in English and Portuguese

Antonio et al.
[TODO: affiliations]

Abstract

[TODO: Write last.]

1 Introduction

[TODO: Motivation; gap; contributions.]

2 Related Work

2.1 External knowledge in NLI

Lexical knowledge has long been used to decide whether a hypothesis word is licensed by the premise. KIM [3] feeds WordNet relations (synonyms, antonyms, hypernyms, co-hyponyms) as pairwise features into the representation, local inference and composition steps of ESIM, yielding gains of 0.6–1.1 points; the ablation shows that ESIM left to learn this alone by a single lexical embedding, precisely the cases where such knowledge should help.

With pretrained Transformers, knowledge is usually injected either as text or as entity embeddings. KRISP [18] appends knowledge-graph triples to the input and restricts their visibility to self-attention, while KnowBert [29] and ERNIE [40] link mentions to knowledge-base entity embeddings.

Graph-based approaches retrieve a subgraph for the pair and encode it with a graph neural network, whose output is fused with the sentence representation [17]. [TODO: add the references of the KGAnon and KGNLI baselines.]

A common limitation is that the retrieved evidence is used as an additional feature vector: it does not change how the decision is reached, and its effect on strong pretrained models is often small. COVER instead lets knowledge drive premise–hypothesis alignment and route the messages that each hypothesis word receives, and the resulting word-level evidence is combined by explicit logical rules.

2.2 Structure inside pretrained encoders

A second line of work integrates graphs with pretrained language models more tightly. DA-GNN [45] reasons jointly over a question-answer context node and a knowledge-graph subgraph, GraphLM [47] exchanges information between the language model and a graph neural network at several layers, and structural adapters [32] insert graph convolutions between frozen Transformer blocks.

These methods target question answering or generation over knowledge or semantic graphs. COVER instead builds its graph over the contextual states of the encoder itself, connecting premise and hypothesis words through lexical relations, and keeps the encoder untouched: the graph operates as a separate aspect whose output is added to the classifier logits.

2.3 Alignment and natural logic

Alignment between premise and hypothesis is a classic ingredient of NLI, from phrase-based adjacency [22] to the soft-attention of the decomposable-attention model [28] and ESIM [2], whose matching function $[\bar{a}; \bar{a}'; a \odot \bar{a}'; a - \bar{a}']$ we reuse in the update of Equation (6).

Natural logic [21] goes further and reasons lexical relations along an alignment, arriving on a set of seven relations $\mathcal{R}$ that we adopt directly. MacCartney and Manning [23] add joint inference, and recent work models the composition problem flexibly [8] or makes it with reinforcement learning and lexical constraints [9].

These systems depend on external aligners or parsers (anyone that are weakly available for English. COVER uses a learned soft alignment with a null target and does not model monotonicity explicitly; the word-to-word relations it discovers it consumes is discussed in Section 4.

2.4 Atomic and probabilistic inference

The aggregation in COVER is chosen to resemble that decide NLI from the parts of the hypothesis. Glöckner et al. [11] defined the probability that a word entails a hypothesis as the product, over hypothesis words, of the probability that each word is entailed by the whole word, an idea later extended by Stacey et al. [35].

In neural NLI, SCIL NLI [35] treats hypothesis spans as logical atoms, predicts a label for each span from sentence-level supervision, and derives the pair label with rules such as "a contradiction requires at least one contradicting span"; it trains its distribution accuracy and improves robustness when training data is scarce.

EPR [43] aligns phrases of the two sentences, predicts a relation for each aligned pair, and combines them with a differentiable fuzzy-logic formulas, and FOLK [20] applies the same principle to facts generated by a large language model.

COVER shares the decision rule of these models (any conflict yields contradiction; full support yields entailment) but differs in three respects. First, its atoms are contextual token states rather than spans, phrases or generated facts, so no chunker or generator is needed. Second, the alignment scores, which starts where a word looks, and as a vector $a(i)$ added to the message, which tells the receiving word what kind of evidence it found (e.g. an antonym rather than a synonym). The bias is initialized to 2 for identical, synonym, hypernym, hyponym, antonym and number relations, 1 for same-stem and 0 otherwise, so that lexical edges are preferred alignments at the start of training and the model can refine them from there.

The update in Equation (6) follows the matching function of ESIM [2] (a_i and a_i' are learned, and learns do not share weights. Unlike pair-wise lexical features [2], the relations here act on contextual states and are refined over two rounds of alignment.

2.2.4 Coverage and conflict head

After the last layer, each hypothesis node $i \in \mathcal{H}$ is scored independently:

$$s_i = \sigma(\mathbf{w}_s^\top \mathbf{z}_i + b_s), \qquad \text{(support)} \quad (7)$$

$$c_i = \sigma(\mathbf{w}_c^\top \mathbf{z}_i + b_c), \qquad \text{(conflict)} \quad (8)$$

$$\alpha_i = \text{softplus}(\mathbf{w}_\alpha^\top \mathbf{z}_i + b_\alpha), \qquad \text{(importance)} \quad (9)$$

s_i is the probability that word i is supported by the premise, c_i the probability that word i contradicts the premise, and α_i how much the word matters to the sentence (a learned scalar, softplus so it is non-negative). Biases are initialized to $b_s = 2$ and $b_c = -2$, i.e. words start as supported and non-conflicting.

2.2.5 Logical aggregation

The word-level scores are combined by two logical rules. The hypothesis is entailed if all of its words are supported (soft-AND in log-space) and is free of conflict if none of its words conflicts (noisy-OR):

$$\log P_{\text{ent}} = \sum_{i \in \mathcal{H}} \alpha_i \log s_i \qquad (10)$$

$$\log P_{\text{con}} = \sum_{i \in \mathcal{H}} \alpha_i \log(1 - c_i). \qquad (11)$$

With $\alpha_i = 1$, Equation (10) is the product of word-level support probabilities of Glöckner et al. [11], and Equation (11) is a noisy-OR over word-level conflicts. The three

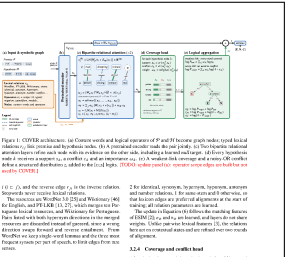


Figure 1: COVER architecture. (a) Content words and logical operators of P and H become graph nodes; typed lexical relations r_{ij} link premise and hypothesis nodes. (b) A pretrained encoder reads the pair jointly. (c) Two bipartite relational attention layers refine each node with its evidence on the other side, including a learned null target. (d) Every hypothesis node i receives a support s_i, a conflict c_i, and an importance α_i. (e) A weighted fuzzy coverage and a noisy-OR conflict define a structured distribution q_ϕ added to the local logits. [TODO: update (used to) / mention seven edges are built but not used by COVER.]

$l \in \{\cdot, \perp\}$, and the reverse edge r_{ji} is the inverse relation. Sequence-wise: token relations.

The scenarios are WordNet 3.0 [25] and Wiktionary [46] for English and OpenWN-PT [1] and PAPEL [31] for Portuguese lexical extraction, and Wiktionary for Portuguese. Pairs listed with both hypernym directions is the merged resources are dismantled instead of guessed, since a wrong direction swaps forward and reverse entailment. From WordNet we keep a high-coverage core of lemmas and the three most frequent synsets per part of speech, to limit edges from rare senses.

2.2.3 Bipartite relational attention

Node states are projected to a smaller space and typed by their sequence.

$$s_i^{(0)} = \text{LN}(\mathbf{W}_p \mathbf{t}_i + \mathbf{E}_{\text{tok}}[\text{id}_i]) \in \mathbb{R}^{d_s} \qquad (2)$$

with $d_s = 256$. Each of $L = 2$ layers lets every premise node attend to the hypothesis nodes and every hypothesis node attend to the premise nodes (cross-sentence links only), thus a lexical node can gather its BPE-word-like a coverage node, or a logical node j to the other side.

$$e_{ij} = \frac{(\mathbf{W}_q \mathbf{s}_i)^\top (\mathbf{W}_k \mathbf{s}_j + b(r_{ij}))}{\sqrt{d_s}}, \qquad (3)$$

$$\alpha_{ij} = \text{softmax}_j\,(e_{ij}, \ldots, e_{il}), \qquad (4)$$

$$\mathbf{m}_i = \sum_j \alpha_{ij} (\mathbf{W}_v \mathbf{s}_j + a(r_{ij})) + a(r_{i\emptyset}), \qquad (5)$$

$$\mathbf{z}_i = \text{LN}(\mathbf{z}_i + \text{MLP}([\mathbf{z}_i; \mathbf{m}_i; \mathbf{z}_i \odot \mathbf{m}_i; \mathbf{z}_i - \mathbf{m}_i])). \qquad (6)$$

The relation enters twice: as a scalar bias $b(r)$ on the alignment score, which starts where a word looks, and as a vector $a(r)$ added to the message, which tells the receiving word what kind of evidence it found (e.g. an antonym rather than a synonym). The bias is initialized to

COVER: Knowledge-Grounded Coverage and Conflict Reasoning for NLI in English and Portuguese

Já escrito

- Trabalhos relacionados
- Metodologia: arquitetura, treino, ablações, baselines, protocolo
- 4 figuras e as tabelas de resultados, geradas automaticamente dos experimentos

A fazer

- Texto dos resultados e discussão
- Introdução, resumo e conclusão

Próximos passos

- Concluir as experimentações (duas máquinas, previsão 14–15/11): ablação PT→EN e baselines de grafo no português e no cross-lingual; K-BERT no inglês.
- Decidir o papel do conflito: manter, tornar opcional ou retirar do modelo principal.
- Corrigir o viés de numerais (`number_conflict`) numa versão 2.
- Artigo: escrever o texto dos resultados; exportar valores reais para a figura de interpretabilidade; calibração (ECE melhor fora da distribuição).
- Pendências do artigo: referências de KGAnet/KGNLI e do InferBR.